

Jessica Tang

Email: jessica.tang@mail.utoronto.ca | Website: jessicatang.github.io

Research areas: Interpretability, AI Safety, Large Language Models (LLMs), Natural Language Processing (NLP)

EDUCATION

University of Toronto

Sept 2022 – May 2026

BASc in Engineering Science (Machine Intelligence)

- *Relevant Courses:* Matrix Algebra & Optimization, Foundations of Computing, Probabilistic Reasoning, Computational Linguistics, Artificial Intelligence, Decision Support Systems

RESEARCH EXPERIENCE

Vector Institute

Sept 2025 – Present

Senior Research Thesis

- *Supervisors:* Prof. Sheila McIlraith and Dr. Silviu Pitis
- *Paper:* **Editing Prompts to Optimize a Margin of Safety in Large Language Models.** Under Review at AAAI 2026 Machine Ethics workshop.
- *Thesis:* Attribution, Interpretability, and Conflict Detection for Behavioural and Safety Alignment in LLMs.
 - Developing a framework for faithful output-to-input attributions in LLMs. Investigating behavioural conflict detection, grounded in faithful attributions for more consistent and safe model behaviour.

Microsoft Research

May 2025 – Present

Undergraduate Research Intern (5% acceptance rate)

- *Mentors:* Dr. Sharad Agrawal and Dr. Shraddha Barke
- *Project:* Traceability in Multi-Agent LLM Systems.
 - Investigating interpretability of multi-turn and multi-agent LLM systems using attention-based influence tracing for debugging and transparency.

KITE Research Institute

Apr 2024 – May 2025

Machine Learning Research Intern

- *Supervisors:* Prof. Shehroz Khan and Dr. Ali Abedi
- *Project:* Rehabilitation Exercise Quality Assessment and Feedback Generation Using Large Language Models
 - Led the end-to-end design of an *LLM-driven feedback system* for rehabilitation exercise assessment.
 - Proposed an adapted Auto-CoT prompting method for physical therapy data.
 - Developed ST-GCN and VQ-VAE pipelines for spatial-temporal skeleton data classification and explainability using Grad-CAM.

Cognitive Neuroscience and Sensorimotor Integration Lab

May 2023 – May 2025

Lab Manager, Computational Neuroscience Research Assistant

- *Supervisor:* Prof. Matthias Niemeier
- *Project:* Task-driven emergence of dorsal and ventral visual streams in artificial neural networks
 - Modelled dorsal and ventral visual stream emergence via task-driven CNNs, achieving >80% accuracy in biologically inspired tasks.
 - Performed explainability analyses (Neuron Shapley, activation maximization) to study neural information flow and graph connectivity.
 - Programmed automated EEG–motion capture synchronization and developed experimental control interfaces in Python.

AWARDS & HONOURS

\$30,000 Award for Diversity and Innovation in Technology, <i>Royal Bank of Canada</i>	2025, 2024
\$7500 Research Award, <i>Transform HF</i>	2024
\$400 Best Project Overall, <i>Cohere</i>	2023
\$2000 Dean's Merit Award, <i>University of Toronto</i>	2022
\$1250 BC Achievement Scholarship	2022
\$1250 District/Authority Scholarship in Technical and Trades Training	2022
\$1000 Ingenious+ National Finalist and Regional Innovation Winner	2022
SGD 1000 Best Presenter, <i>International Student Conference On Artificial Intelligence</i>	2021
\$560 AI4Impact Grant Award, <i>AI4ALL</i>	2021

PUBLICATIONS & POSTERS

- Measuring and Mitigating Instructional Conflicts in Large Language Models
J. Tang, S. Pitis, S. McIlraith. *In Preparation*.
- Rehabilitation Exercise Quality Assessment and Feedback Generation Using Large Language Models with Prompt Engineering.
J. Tang, A. Abedi, T. S. Colella, S. Khan. *IJCAI ARIAL Workshop, 2025. Springer Nature, 2025*.
- Emergence of dorsal-like and ventral-like properties in artificial neural networks (Poster)
T. Reza, S. Luo, E. Jordan, **J. Tang**, K. Patel, M. Niemeier. *Society of Neuroscience Conference, 2024*.
- Comparative analysis of optimization trends in dorsal and ventral stream using computational model (Poster)
T. Reza, S. Luo, G. Singh, **J. Tang**, R. Jain, M. Niemeier. Poster. *Cognitive Neuroscience Society Conference, 2024*.
- Modular Can-Sized Satellite System with Active Attitude Control (Outstanding Paper Award)
T. Cai, K. Howard, B. Zhou, **J. Tang**, V. He, D. Liu. *European Space Agency International CanSat Competition, 2022*.
- Deep Reinforcement Learning Controller for Indoor Farming (Best Presenter Award)
Jessica Tang. *International Student Conference on Artificial Intelligence, 2021*.

LEADERSHIP, TEACHING

- President and Founder, *IlluminAI*** **Jul 2020 – Present**
- Founded an organization increasing AI ethics education; reached 1,000+ participants across 11 countries.
 - Directed 16 interdisciplinary workshops and events with speakers from Zoom, IBM, MIT, and UofT.
 - Organized the **UofT AI Ethics Hackathon**, engaging 15+ disciplines across technical and non-technical fields.
 - Secured \$3,000+ in funding from AI4ALL, UofT, NCWIT, SAP, and Perplexity AI.
- Creator and Editor, *YouTube Channel*** **Aug 2023 – Present**
- Built an audience of **~3.4K subscribers** and **~1.17K views**, sharing insights on engineering, research, and women in STEM.
 - Combined **music, storytelling, and research communication** to raise awareness for diversity and inclusion in technology.
- Course Instructor, *Wave Learning Festival*** **June 2021 – Jul 2021**
- Designed and taught 'Introduction to Artificial Intelligence' to 70+ students, covering neural networks and AI ethics.
- AI Scholar, Curriculum Developer, *AI4ALL*** **Jul 2020 – Apr 2021**
- Selected among 32 students nationwide; developed a CNN for facial expression classification.
 - Created AI curriculum later adopted by Bronx School of Science, featured by NCWIT.